Carlos Romero Madrid

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Computational Physicist • HPC Software Engineer • Creative Technologist

Profile & Mission: Computational Physicist (Double Degree Physics & Mathematics, 1st of Class MSc Plasma Physics) and HPC Software Engineer at BSC with 5+ years experience. Bridging extreme-scale HPC clusters (MN5, Leonardo, LUMI) with physical computing, generative sound, and narrative data visualization for the **democratization of science**. Actively pairing with LLMs for high-velocity cross-disciplinary prototyping.

BSC HPC Engineer MN5, ClimateDT & DestinationE	**Physics & Mathematics** Double Degree & Plasma MSc (1st)	**Physical Computing** ESP32, 6-DoF IMU & I2S Sonification	**Visual Storytelling** Expo-Curator, Photo & After Effects

Featured Creative Computing & Data Projects

MNusic — Visualizing & Listening to MareNostrum 5 & LUMI

HPC DATA SONIFICATION

• Synaesthetic telemetry engine translating tier-0 supercomputing metrics into real-time harmonic soundscapes (Web Audio API) and generative dynamic particle visuals.
• Maps compute utilization, GPU tensor spikes, interconnect latency, and cooling thermal gradients into microtonal frequencies and acoustic harmonics, humanizing petascale computation.

Pipa-Music-Ball — Non-Human Bio-Kinetic Interactive Sound System

PHYSICAL COMPUTING & SENSORY IOT

• Embedded microcontroller-powered physical sphere translating feline kinetic play (velocity, tilt, angular acceleration) into generative microtonal acoustic phrases.
• Places animal curiosity at the heart of the generative loop; tangible proof that data interfaces can be playful, tactile, and non-anthropocentric (ESP32, 6-DoF IMU, I2S DAC).

Expo-Curator — Visual Corpus Analytics & Exhibition Staging

CORPUS ANALYTICS • GITHUB

• Open-source studio application for image curation, automated sub-classification, booklet design, and exhibition wall staging.
• Streamlines spatial narrative organization and print-ready editorial catalogs for cultural spaces, science centers, and multidisciplinary exhibitions.

After Effects Lab — Procedural Motion Design & Kinetic Animation

VISUAL MEDIA & MOTION DESIGN

• Creative investigation combining still photographic framing, spatial composition, and lighting with procedural Adobe After Effects motion graphics.
• Translates abstract scientific telemetry into broadcast-quality cinematic videos, kinetic typography, and exhibition reels.

Technical Matrix & Creative Capabilities

Data Art & Visualization	Motion & Visual Media	AI, Vision & Physical Computing
• Tangible Data Interfaces	• **Adobe After Effects** (Motion/AE)	• **Computer Vision & MediaPipe**
• Web Audio API Sonification	• **Still Photography** & Framing	• PyTorch, Embeddings & Latent Space
• HTML5 Canvas & Generative Art	• **Expo-Curator** Exhibition Studio	• Physical Computing (ESP32, 6-DoF IMU)
• Narrative Architecture & Infographics	• Video Rhythm & Kinetic Typography	• CUDA GPU & Slurm Cluster Scaling
• Inspired by Domestic Data Streamers	• Color Grading & Visual Systems	• **Agentic LLM Pipelines & Web ML**

Professional Trajectory & Academic Foundation

Barcelona Supercomputing Center (BSC-CNS) — HPC Scientific Software Engineer

May 2023 — Present

• *Performance Team (May 2025 – Present):* Ocean-atmosphere coupling in DestinationE & ClimateDT digital twins; developing ECLand model; deploying workflows on MareNostrum 5.
• *Fusion Team (May 2023 – Apr 2025):* Optimized nuclear fusion simulation codes with OpenMP/MPI/CUDA/OpenACC; implemented superconducting magnet modeling in ALYA-MAGNET.

National Renewable Energy Centre (CENER) — Data Scientist

May 2022 — Sep 2022

• Built Python real-time telemetry data pipelines for characterisation and predictive maintenance of offshore wind plants using Pandas and Scikit-Learn.

Univ. of Seville (Research Scientist) & Fraunhofer FHR (Cognitive Radar) 2018 — 2022

• Particle-in-cell (PIC) & MHD plasma simulations on HPC systems; lectured Quantum Physics; MATLAB simulated annealing optimization for non-planar radar antenna design.

Education: Double BSc Physics & Mathematics (8.35 · 3rd) • MSc Plasma (15.43 · 1st) Univ. Seville • Aix-Marseille • UNED

• Double BSc in Physics & Mathematics (Univ. of Seville, 2013-2018); MSc Plasma & Fusion Physics (Aix-Marseille Univ., 2019-2020); MSc Advanced Mathematics (UNED, 2023-Present, ABD).